



Foundations of Cybersecurity

CyberSec is the practice of ensuring confidentiality, integrity, and availability of information by protecting networks, devices, people, and data from unauthorized access or criminal exploitation.

Threat actor: any person or group who presents a security risk

Compliance: Internal standards and external regulations to avoid fines

Security Framework: guide line for making plans to help reduce risks.

Security Controls: safeguards to reduce specific security risks. Use Security Framework to establish strong security posture.

Security posture: orgs ability to manage its defense of critical data.

internal threat: current or former employee, who is a risk to security.

network security: process of keeping an orgs data, services, systems, and devices safe.

cloud security: ensure that assets are safe and stored in the cloud.

CIA triad: a model that helps organizations consider risk when setting up systems and security policies.

Confidentiality, Integrity, Availability

programming: process to give a computer instructions to perform tasks.

PII (**Personally identifiable information**): info used to infer an individuals identity

SPII (**Sensitive personally identifiable information**): PII that have stricter guide lines

Malware: Software made to harm devices or networks

Computer Virus: Malicious code written to change computer operations

Worm: self replicates and spreads across the network

Spyware: install malware without users permission

Ransomware: demand payment after encrpyting orgs data

Phishing: using digital communication to trick people into giving up data

- spear phising: email attack that targets a specific trusted users or groups

- whaling: target executives

- BEC (Business email comprise): threat actor impersonates a known scource for financial gain

- vishing: exploit Electronic voice communication for data

Social Engineering: Manipulation technique that exploits human error

- Social media phising: threat actor collects detailed info about their target

- watering hole attack: threat actor compromises a website frequently visited group of users

- Physical social engineering: threat actor impersonates employee, customer, vendor for location

- USB baiting: threat actor leave a malware USb stick for an employee